



Virtual reality in the automotive field in industry 4.0

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ABSTRACT

Virtual Reality (VR) and Augmented Reality (AR) are the new computer technologies to create a simulated environment in the field of automotive. Virtual Reality can create for its users an experience that simulates the real environment in the field of automotive and creates a 3D image. Instead of having a screen in front of them, users interact with 3D world. By using VR and AR many senses can be stimulated, such as vision, hearing, touch, even smell and noise, the computer being transformed into a gatekeeper to this 3D artificial world.

The paper will present a state of art on the needs of the Virtual Reality and Augmented Reality devices used in the automotive field and two case studies related to some applications of Virtual Reality in the automotive field in Romania.

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1. Introduction

Virtual development started in early 1970's, at first with a CAD software in the design field.

At the beginning of 1990 the first public demonstrations of Virtual Reality (VR) started to take place in exhibitions and the outlook was quite promising for some of the observers from industry which started their own experiments.

Volkswagen began to acknowledge the importance of Virtual Reality (VR) and Augmented Reality (AR) in 1994, after a visit at the New York Virtual Reality exhibition and Silicon Valley Virtual Reality exhibition. First, they have made a research regarding some software and hardware. For software, they started a cooperation with the Fraunhofer Institute in Darmstadt with Prof. Encarnacao. The next step was to buy a Silicon Graphics computer and some other devices, including space mouse, data glove and HMD, but nowadays the field of VR is very strong developed [12].

VR and AR technology are very important in Industry 4.0, the combination of these technologies can improve the methods and efficiency of the companies in manufacturing field. The Industry 4.0 brings some new technological challenges in the process that gradually made people adapt and increase the level of the digitalisation.

VR and AR technology are used in the industry, in professional training (military, medical, etc.), in education, simulations, evaluation of designed models, studies related to product ergonomics, rapid prototyping, some example are in Fig. 1.

In the automotive industry, VR and AR technology have proven to be a useful tool also in assisting users in product evaluation processes and certification of the process.

In 1999, BMW started to explore and use VR's ability to analyse products in the design phase. The results demonstrate that VR technology has the potential to reduce the number of required versions of the model. Motorola also developed in 1995 a VR system used to train employees in an assembly line. The conclusion of the experiment showed that VR can be used successfully in professional training, with good results in comparison to classical training.

Boeing, the world's largest aircraft manufacturer, has developed the Virtual Space eXperiment (VSX) based on VR/AR technologies. This product is a demonstration of the usefulness of VR technologies in the aerospace industry and other complex industries involving machine-user interaction, the product can be seen in Fig. 2.

VR and AR in automotive have the potential to reduce the number of screens inside the car and improve their design, production, and maintenance, as well as the driver's experience. This technology has many applications in the automotive industry. One of these is assisting mechanics in their tasks.

AR creates an extension of the three-dimensional environment through virtual objects. This technology can provide important

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Fig. 1. Virtual reality glasses: (a) VR glasses used in automotive, (b) VR Samsung glasses, (c) VR glasses used for inspection [17].



Fig. 2. (a) Boeing 737 Flight Simulator inside, (b) Boeing 737 Flight Simulator outside [13].

assistance in traffic, where distances and location in space are the most important factors for road safety.

Another product that use AR technology is ADAS (Advanced Driver Assistance Systems), this product aims to improve the driving experience by monitoring the external factors (Fig. 3). Positioning and spatial relationships are evaluated and possible future events can be calculated. This system warns the driver if the time left until the event will occur is enough for him to react. Also, some tasks such as braking can be automated, thus obtaining a faster response time to external factors [6,8-11].

Automotive companies are in a continuous race to reduce the time of development and delivery of products, while seeking to reduce costs and increase quality, in other words to provide smart products. Using the new technologies these companies have integrated AR into the Product Lifecycle Management (PLM). In this cycle the stages can be succeeded more easily due to these new technologies.

VR solutions applied in roadside assistance want to reduce the response in time to external information and increase the safety in driving. The only limits to near-real VR experiences are the availability of content and the performance of the devices used.

Some statistics from the field of automotive shows that the number of virtual, augmented and mixed reality headset devices will increase very much between 2015 and 2023 [2,3,5].

2. Benefits of AR and VR training in automotive and aircraft manufacturing

The training with VR and AR technologies in the industry can give a completely new perspective of preparing employees and processes. Workers participating in the training, due to the imposition of virtual models on real images using AR or VR technologies and devices, can see exactly, for example, what steps they should take in order to replace a damaged component or disassemble

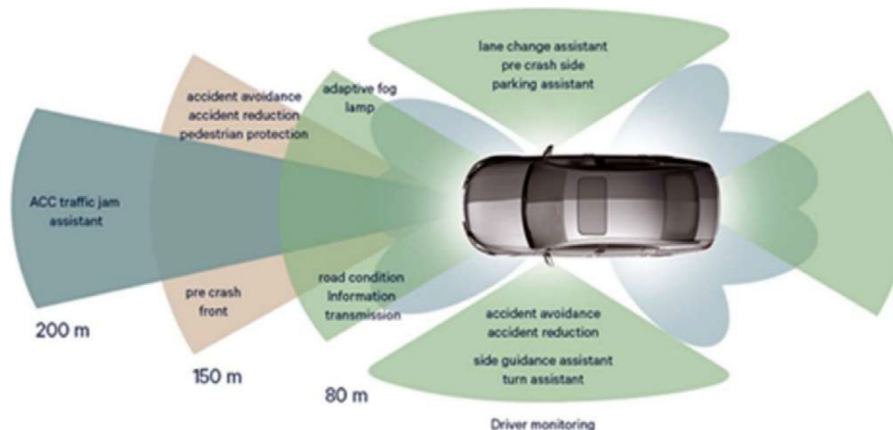


Fig. 3. ADAS Advanced Driver Assistance Systems [14].

the machine components. All they need to do is to follow the steps from the virtual training. These techniques save time and money, compared to regular training and tests. Companies such as Volvo, Ford or Hyundai, use VR / AR to increase their efficiency in the design phase, processes phase as well as in marketing or sales [2].

The recent studies suggest that with the employees who started to work with VR tools (were equipped with a tablet), the company found a significant improvement of 90% in the quality of execution. The engineers who started to work with VR tools have accelerated the speed of assembling parts of the aircraft by 30%. It means, within the same time, that a team can assemble 1.3 times the normal speed [7].

In the same contexts of the recent studies, the workers in Boeing have shown a significant development while using smart glasses with VR and AR technologies. The technicians used the smart glasses with VR and AR technologies to cut fine wires and the production time increased by 25%. It is very clear how the application of VR, AR, and Mixed Reality can enhance the degree of production in the manufacturing of airplanes and how the cost can be adjusted by using this technology [14].

SOLDAMATIC Augmented Training is the first welding training solution powered with AR in the world. It applies new technologies like AR to enhance the learning process and uses real welding equipment to develop skills and muscle memory. Soldamatic can be used in aircraft industry and in the automotive field for training. Seabery with their Soldamatic (Fig. 4) product, want to become a reference in the educational sector, they want to change the traditional educational model to a more personalized one applying the concept of Augmented Training. This method includes AR as an added value to enhance the teaching–learning process. The simulator can improve productivity, reduce failure in recertification, because the process can be simulated before production on the simulator [16] Figs. 5–8.

In Europe, Augmented Training was being adopted and implemented by welding training centers, societies and big industries like Siemens, Daimler Mercedes, Saarlühl, Volkswagen, Man, Liebherr or John Deere, among others, to improve efficiency and safety.

For welding skills using Soldamatic's AR, the trainees will have several substantial benefits, such as:

- reduce the quantities of materials – 68% average reduction of training materials
- flexible training times for students – faster students can move ahead when ready, others could take more time if they needed, resulting in a 37% greater throughput of students and a 17% overall reduction in training time
- 34% increase of the percentage of trainees who qualified for certification
- 17% reduction of the cost of welding instructors
- 84% reduction in accidents, which can be very serious in welding training



Fig. 4. (a) Virtual training in welding, (b) Virtual training in production [16].

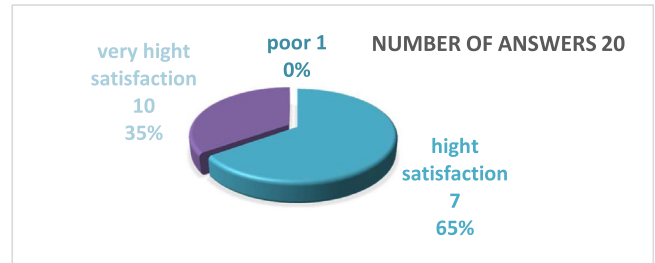


Fig. 5. Number of answers of questionnaire [19].

- high student satisfaction evaluation scores – from a list of 20 evaluation questions rated on a 0–10 scale, 65% of responses were 7 or above, and 35% were 10

Seabery declared that since 2014, over 1,500 Soldamatic units have been sold in more than 40 countries. The technology has been used in training by some of the world's largest manufacturers, including Siemens, John Deere, and Volkswagen, in addition to uptake by both international educational institutions and government bodies.

Soldamatic had conducted tests comparing their Augmented Reality technology to traditional welding training. The results demonstrated that 34% more welders were certified in 56% less time, saving up to 68% on the overall cost of welder training. In addition, Soldamatic increases the time on arc by three to five times and enables training institutes to educate four times more students while maintaining their existing lab infrastructure [18].

VR/AR enables the trainee to get hands-on experience with real-time industrial problems by practicing them multiple times, without the need to visit the plant in real. The trainee practices to instinctively respond to situations & retain the action performed in the form of muscle memory. The continuous practice on all possible critical situations in VR/AR will help in building the trainee's muscle memory & amplifies the pace of learning.

3. Case study 1– LED light system for automotive business

This case study presents a new product development example, where VR technology integrates the design process with customer collaboration to increase the product performance and quality.

The new product development supposes a lot of resources to be involved, starting with definition of customer or market requirements, deep engineering investigation and good collaboration between stakeholders.

The new product development in Industry 4.0 concept, means smart product consisting of mechanical, electronics and very often software integration, which requires a multi-discipline team work for a common purpose.



Fig. 6. Night drive LED light simulation: (a) VR glasses, (b) road simulation [17].



Fig. 7. VR training Linde [15].

The product performance and quality depends on clear requirements definition and integration of requirements during product development phases.

In traditional product development approach, the integration of customer requirements means a lot of iteration during design phase, which increases the time and development cost. As a consequence, the product becomes either too expensive for the market or is released too late on the market.

Industry 4.0 proposes a set of digital and smart technologies that integrates complete product life cycle, reduces product development phase, increases flexibility and performance on affordable costs.

VR is one of the technologies associated with Industry 4.0, which simulates the real world on computer based and supports the product development process without physical environment.

As a case study, it was proposed for design, a new LED auxiliary lamp for a car, with the main purpose to increase the light output during the night drive.

The customers desire is translated by engineers into technical parameter that need to be fulfill, based on similar products on

the market and the new product is required to achieve minimum 3000 lm.

Once the customer expectation is defined, the optical design engineers start to calculate the number of LEDs and perform computer simulation to reach the required light output. At the same time, the electronics engineers must define the electronic concept and determine the system behavior. One of the most critical dependencies between LEDs and electronic concept is thermal behaviour. While the heat increases into the lamp, the LEDs performance is reduce, leading to light output being insufficient.

In top of all, there are legal regulation and safety requirements, which consist of defined light range for each source of light and for each region. In addition, each standard clearly states which lamp is mandatory by law and which are optional.

As already mentioned, the product development is an iteration process, especially when high performance requirements are on the technology edge.

In such a situation, customer should be part of process development and understand the technical, safety and legal requirements, in a collaborative platform.

Beside early customer involvement into design process and decision, the cost for prototyping and iteration on product optimization according with customer needs and legal aspects, is significantly reduced.

The VR technology is the right digital tool as it can transfer the engineering parameters into visual computer simulation for different design options and customer can be involved in deciding the most convenient appearance for final product itself.

By simulating the driving experience over the night, on one hand improves the visibility on different angles, on the other hand considers the safety requirements for other participants, especially in case of auxiliary light source.

4. Case study 2- LindeGO smart glasses

Linde company had developed using VR and AR technology, some applications that can be used to train operators before a plant

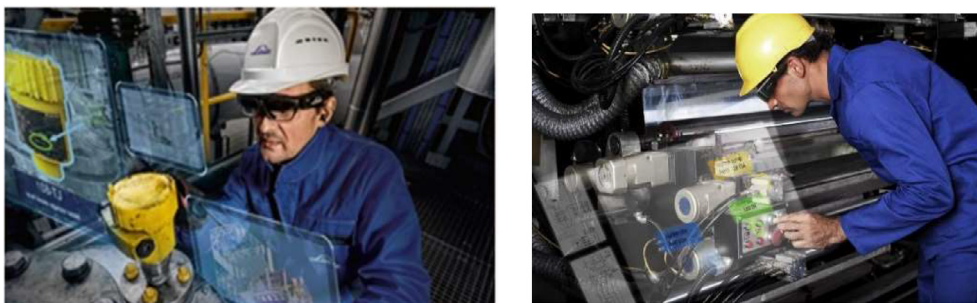


Fig. 8. LindeGO smart glasses [15].

has even been finished, applications as LindeGo smart glasses, FreezAR App or the application for a virtual tour on a plant.

This year in München (2020), Linde used the INNOACTIVE PORTAL to launch the Linde Virtual Academy, a continuously growing library of VR training apps for plant operators in the oil & gas industry. By using this library of VR training, Linde wants to increase the flexibility of their products and services.

In Romania, LINDE uses some applications with AR and VR, two of them can be mentioned: LindeGO smart glasses and FreezAR App for the remote support in the different plants. When using LindeGO smart glasses application, an instant live audio-visual link can be arranged with one of Linde experts, who can then see the plant live and can try to solve the problem. The on-site technician can immediately follow the information and instructions over the live audio-visual connection.

When industrial processes at a large plant come to a halt, solving the problem can stretch the expertise of in-house technicians or engineers and the problem cannot be solved locally. Using the application LindeGO smart glasses, the experts can provide immediately a solution or a recommendation. When it is a critical issue, the plant needs immediate expert support – ideally from the contractor who built the plant and knows all the inside and outside site layout. If the plant is in a remote location, however it can take days for specialists to get to the site. All of that means undesirable delays and loss of earnings. In our days, with VR/AR digital technologies from LINDE, this problem was solved. With the VR applications, Linde can offer immediate support online everywhere in the world.

LindeGO smart glasses developed in AR are helping to digitally transform the working culture at LINDE by enabling instant access to expertise – whenever and wherever is required in the world.

LindeGO smart glasses is just one device in Linde's remote support solution portfolio, this product needs specially adapted smartphones and tablets. These devices will be connected with the glasses, wearable devices have been integrated audio and video, which means remote experts can connect via a live stream and see what the on-site operator is seeing as they tackle a task and can see the environment from all the angles. The device can offer all the expert support and feedback need in real time. Then, the expert can send digital information in real-time pointers, drawings, site plans and photos which are juxtaposed with the physical world within the wearer's field of view [15].

Both applications (LindeGO smart glasses and FreezAR) are used in the plants in automotive field in Romania. LINDE currently collaborates with: Ford, Renault, Bosh, Emerson, etc.

LindeGO remote application service is also very friendly and the user has only a few steps to follow for installation:

- search the App LindeGO in the App Store or Google Play
- download the app and install
- register for the app by requesting your login details via the app contact form or by sending an email to PLANTSERV@LINDE-LE.COM (name and surname, email, username, company)
- for support, a request can be send to your LINDE PLANTSERV® contact
- the user will liaise with the right expert(s) and align the timing with you
- simply dial into the video session at the agreed time and the Linde expert will provide you with live support, possibly also using your smartphone camera function to get a real-time view of the plant

With the help of AR/VR devices (LindeGO) the company can expand cooperation with the local community contributing to the fulfillment of their needs (in the field of monitoring and security). AR/VR devices can be used to detect wastage in manufactur-

ing processes, which can have a positive effect on reducing pollution, thus reducing costs. The world technological progress and the development of digitization currently provide modern solutions that can be implemented in the field of sustainable development of enterprises, which is a part of the field of management science.

5. Conclusions

This paper presents a state of the art of the device used on VR and AR in automotive and two case studies in the fields of VR and AR in automotive in Romania.

Digitalization had made good progress on product development in the automotive industry in the last period and especially due to the recent pandemic, from this reason all the fields needs to increase the level of the digitalization and a lot of tasks now are made online and virtually.

Analyzing the two case studies it can be noticed that by using VR and AR technologies the companies can be cover a lot of tasks from the engineering fields, for example the control in plants (LindeGo application).

Another important step in automotive field is the training field and in the case of training using AR and VR technologies the company can reduce the cost of the training courses and due to the recent pandemic the training activity can be made, the trainees don't need to be face to face, the activity can be done from distance.

AR or VR technologies and tools are quite effective and useful in managing corporate business processes, in first stages at HR (for the purposes of recruiting, training, development, personnel evaluation) and basic operational processes (3D-visualization, modeling of production processes, large-scale projects, elimination of "weak" places, automation, modeling of IT risks, cyber security, etc.). The development of the VR/AR industry and the change of the focus from the entertainment sector to the corporate sector are evidenced by research data of international companies and statistical organizations, including PwC, IDC, Statista, etc. The growth in consumption of Internet technologies, the export of IT and mobile products, and the growth of mobile technology markets are indicative of the prospects for the development of AR or VR devices and field.

The cases studies presented in the paper demonstrate that the VR or AR technologies are very important in the corporate business processes, the first case study (Night drive LED) can reduce the cost for prototyping and iteration for product optimization according with customer need and legal aspects and the second can offer maintenance at the large plants, this study can be continued in the idea of analyzing the sustainability and the impact of the devices presented.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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